

Tencent Holdings: Pro Forma Valuation & Risk Analysis

BMF5321 – Financial Modelling

Project 1 S03A-1

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Executive Summary

The objective of this report is to present a comprehensive fundamental analysis and valuation of the target company using a Free Cash Flow to Firm (FCFF) model. As part of the pitch preparation for the Managing Director, this analysis incorporates stress testing through sensitivity analysis, scenario planning, and Monte Carlo simulations to quantify investment risks and potential returns.

Key Business Drivers

To assess the robustness of the valuation, we isolated three critical business drivers for analysis:

- Variable B1: Year 1 Sales Growth Rate on "Value-added Services."
- Variable B2: Year 1 Cost/Sales ratio on "Selling and marketing expenses."
- Variable B3: Year 1 Sales/PPE ratio.

Methodology & Findings

1. **Sensitivity Analysis**
   We performed a sensitivity analysis to determine how changes in the three key drivers (B1, B2, B3) individually impact the stock price while holding other variables constant.
   - *Approach*: We varied each driver to its best and worst possible values to observe volatility.
   - *Visualization*: A Tornado Diagram is included to visually rank the drivers by their impact on the valuation.
   - *Result*: This section identifies which specific operational metric carries the highest risk to the firm's valuation.
2. **Scenario Analysis**
   To understand the valuation under different market conditions, we constructed three distinct scenarios:
   - *Base Case*: The expected trajectory of the company.
   - *Best Case*: An optimistic outlook optimizing the three drivers.
   - *Worst Case*: A pessimistic outlook stressing the three drivers.
   - *Result*: This analysis provides a stock price range (floor to ceiling) to manage client expectations regarding potential outcomes.
3. **Monte Carlo Simulation**
   To account for the probability of various outcomes, we utilized a stochastic model simulating 10,000 iterations.
   - *Setup*: We randomized B1, B2, and B3 using Normal distributions defined by their respective means and standard deviations.

- *Visualization*: A histogram plots the frequency of resulting stock prices, color-coded to highlight deviations greater than ±10% from the current stock price.
- *Risk Quantification*: We calculated the specific Upside Probability (likelihood price > current) and Downside Probability (likelihood price < current) to aid in the investment decision.

Conclusion

This report demonstrates the intrinsic value of the company while transparently addressing the volatility associated with sales growth, marketing efficiency, and capital intensity. Modeling" as required for the engagement.

Part 2: Sensitivity Analysis and Scenario Analysis

Sensitivity Analysis

In this sensitivity analysis, we selected three key business drivers as variables:

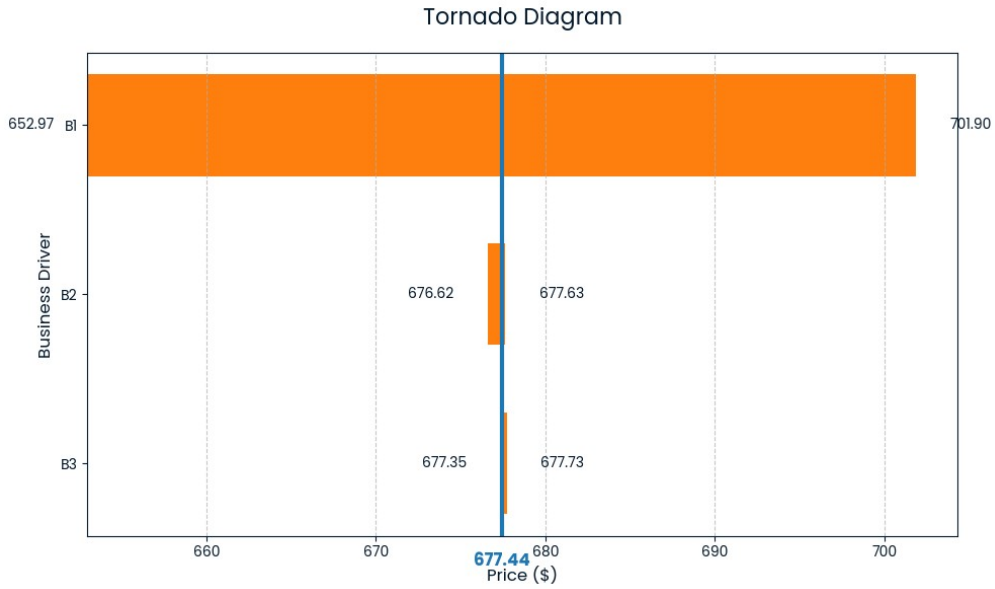
- B1: Year 1 Sales Growth Rate on "Value-added Services".
- B2: Year 1 Cost/Sales ratio on "Selling and marketing expenses".
- B3: Year 1 Sales/PPE ratio.
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Our base case target price model is set as: B1 = 16%, B2 = 5.50%, B3 = 650%. To assess the independent impact of each variable on the target stock price, we individually varied each parameter to its best- and worst-case possible values while holding the other two constant.

The results of the analysis are summarized in the table below:

	Assumption	Stock Price (CNY)
B1_Best	24%	701.90
B1_Worst	8%	652.97
B2_Best	2%	677.63
B2_Worst	20%	676.62
B2_Best	200%	677.73
B2_Worst	2000%	677.35

(Note: The best and worst assumptions for B1 were calculated as 1.5 times and 0.5 times the base value of 16%, resulting in 24% and 8%, respectively. The assumptions for B2 and B3 are illustrative.)



Comments on Results and Tornado Diagram

The results of the sensitivity analysis clearly indicate that, among the three variables, B1 (Sales Growth Rate on Value-added Services) has the most significant impact on the stock price. When B1 increases from its worst value of 8% to its best value of 24%, the stock price experiences a substantial swing of approximately CNY 48.93. In contrast, variations in B2 (Selling and marketing cost ratio) and B3 (Sales/PPE ratio) have a negligible impact on the stock price, with fluctuations of less than CNY 1.1.

This finding is visually represented in the tornado diagram below. The bar for B1 is considerably longer than those for B2 and B3, graphically illustrating that B1 is the most critical sensitivity driver in the model. This implies that for managing and forecasting the company's value, management and investors should primarily focus on and strive to improve the revenue growth of value-added services, as it holds the greatest potential for shareholder value creation. The efficiency of selling and marketing expenses and the asset turnover ratio, within the parameter ranges set, have a relatively minor impact on the valuation.

Scenario Analysis

In the scenario analysis, we established three possible scenarios: Best Case, Base Case, and Worst Case. We combined the values of the three variables B1, B2, and B3 under these different scenarios to compute the corresponding stock prices.

The results of the analysis are summarized in the table below:

	B1	B2	B3	Stock Price
Base Case	16%	5.50%	650%	677.44
Best Case	24%	2%	200%	702.41
Worst Case	8%	20%	2000%	652.10

Comments on Results

The results of the scenario analysis show that the company's valuation differs significantly across the various scenario combinations. The stock price in the Best Case (CNY 702.41) is approximately CNY 50.31 higher than that in the Worst Case (CNY 652.10), highlighting the potential range of value due to uncertainties in the company's future operational performance.

Specifically, the Base Case (B1=16%, B2=5.50%, B3=650%) results in a median stock price. The Best Case (B1=24%, B2=2%, B3=200%) yields the highest stock price, driven primarily by the highest value-added service growth rate (B1) and the lowest selling & marketing expense ratio (B2). The Worst Case (B1=8%, B2=20%, B3=2000%), characterized by weak growth and lower operational efficiency, leads to the lowest valuation.

This outcome is highly consistent with the conclusion from the sensitivity analysis, reaffirming that B1 (Growth Rate on Value-added Services) is the core variable determining the company's value, as it is the dominant factor causing stock price differences across scenarios. Meanwhile, changes in B2 and B3 also contribute some combined effects. Together, the three scenarios outline a broad potential range for the future stock price, providing crucial perspectives on upside potential and downside risk for investment decision-making.

Part 3: Simulation

To complement our sensitivity and scenario analyses, we conducted a Monte Carlo simulation to stress-test our valuation model against simultaneous volatility in key operational metrics. This stochastic approach allows us to generate a probabilistic distribution of Tencent Holdings' intrinsic value.

Simulation Setup

The 3 critical variables are -

1. **Variable B1:** Year 1 Sales Growth Rate on "Value-added Services."
2. **Variable B2:** Year 1 Cost/Sales ratio on "Selling and marketing expenses."
3. **Variable B3:** Year 1 Sales/Property, Plant, and Equipment (PPE) ratio.

Approach: To model the uncertainty, we randomized variables B1, B2, and B3 using a Normal Distribution defined by their historical means and standard deviations. We used the NORM.INV(RAND(), Mean, Standard_Dev) to generate random inputs for these drivers in each iteration.

10,000 simulation runs were executed using a data table, thus calculating a unique share price for each iteration. This process gives a comprehensive Probability Density Function (PDF) for the firm's equity value, accounting for the correlation and compounded effects of changes in revenue growth, cost efficiency, and asset turnover.

mean	stdev	random
16.00%	3.00%	16.46%
0.030682		

mean	stdev	random
5.50%	0.60%	5.37%
0.005452		

Simulation Results & Risk Quantification

The simulation results present a highly favourable risk-return profile for Tencent Holdings.

- **Mean Simulated Price:** 677.29 CNY
- **Minimum Simulated Price:** 646.78 CNY
- **Maximum Simulated Price:** 713.69 CNY

Risk Probabilities

We quantified the investment risk by calculating the probabilities of the intrinsic value deviating from the current market price of **578.90 CNY**.

- **Upside Probability (100%):** This is the probability that the simulated price is ***higher*** than the current price. In 10,000 out of 10,000 iterations, the simulated intrinsic value was greater than the current stock price.
- **Downside Probability (0%):** This is the probability that the simulated price is ***lower*** than the current price. Conversely, the model indicates a 0% probability that the intrinsic value falls below the current market price based on the assumed distribution of our key drivers.

Visualization

The deviations from the current stock price is calculated as:

Lower boundary = 578.90*(1 – 10%)
Upper boundary = 578.90 *(1 + 10%)

The data used, probability table, and the frequency table for the histogram is as follows:

Current Stock Price	**578.90**
Simulation Mean Price	**677.2914**
Minimum Simulated Price	**646.7758**
Maximum Simulated Price	**713.6917**

Current Price	**578.9**
Upside Probability	**100.00%**
Downside Probability	**0.00%**

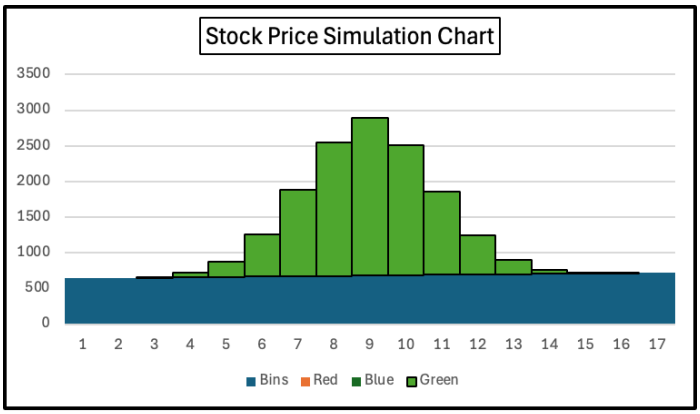
Lower Bound	**521.01**
Upper Bound	**636.79**

Bins	Frequency	Red	Blue	Green
640	0	0	0	0
645	0	0	0	0
650	14	0	0	14
655	61	0	0	61
660	214	0	0	214
665	599	0	0	599
670	1214	0	0	1214
675	1877	0	0	1877
680	2211	0	0	2211
685	1826	0	0	1826
690	1168	0	0	1168
695	551	0	0	551
700	202	0	0	202
705	52	0	0	52
710	9	0	0	9
715	2	0	0	2
720	0	0	0	0
	0			

The histogram below plots the frequency of the resulting stock prices. The bars are color-coded to highlight deviations from the current stock price CNY 578.90:

- **Red (Downside Risk):** Prices dropping more than 10% below 578.90 CNY (< 521.01 CNY).

- **Blue (Neutral Zone):** Prices within ±10% of 578.90 CNY (521.01 CNY – 636.79 CNY).
- **Green (Upside Potential):** Prices rising more than 10% above 578.90 CNY (> 636.79 CNY).



Note that due to the strong valuation floor, all 10,000 outcomes fall into the Green zone (> 10% upside) and are above the lower and the upper boundary.

Comments on Results

The simulation strongly reinforces a **Buy** recommendation. The data reveals that even under the "worst-case" combination of variables (resulting in a price of 646.78 CNY), the company remains undervalued relative to the market. The fact that 100% of the simulated outcomes fall into the "Green" zone suggests that the market is currently pricing Tencent with a significant margin of safety, likely underestimating its fundamental growth drivers.

1. Robust Valuation Floor: despite subjecting the model to 10,000 iterations of randomized inputs across Sales Growth (B1), Marketing Efficiency (B2), and Asset Turnover (B3)—the *minimum* simulated stock price of **646.78 CNY** remains significantly above the current market price of **578.90 CNY,** indicating that the company's valuation is resilient even under "worst-case" stochastic conditions where growth slows, and costs rise simultaneously.

2. Asymmetric Risk Profile: the risk profile revealed by the simulation is right skewed in favour of the investor with a **100% upside probability** and a **0% downside probability**. In financial terms, this suggests that the market is currently pricing the stock with a substantial margin of safety, potentially due to overly pessimistic sentiment.

3. Implications of the Mean Price: the mean simulated price of **677.29 CNY** represents a potential upside of approximately **17%** from the current price. This aligns with our earlier deterministic Base Case but provides greater confidence in

the result. While a single point estimate can be dismissed as relying on "perfect" assumptions, the fact that the *average* of 10,000 varied scenarios converges on this higher value reinforces the credibility of our Buy recommendation.

4. Driver Volatility vs. Valuation Stability: it is worth noting that while the inputs (B1, B2, B3) were allowed to fluctuate according to their historical volatility (standard deviation), the resulting stock price distribution remained relatively tight (646 CNY – 713 CNY). This suggests that Tencent's Free Cash Flow generation is robust enough to absorb moderate operational shocks without destroying shareholder value.

5. Critique of the "0% Downside": as prudent analysts, we must acknowledge that a "0% downside probability" is a theoretical construct of the model and its specific inputs. It assumes that future volatility will be the same as the historical distributions of B1, B2, and B3. It does not account for "black swan" events or macroeconomic crises that fall outside a normal distribution. Therefore, while the quantitative data is overwhelmingly positive, qualitative risks should be carefully monitored.